2026-03-22 - notatki SWJ

Stacja pogodowa APRS przy wykorzystaniu urządzenia Hydrawise Garni 2055 ARCUS

Stacja pogodowa po testach

Przetestowany pliki i procedura co po kolei

Na bazie projektów:

https://github.com/ArturVSS/APRS-WX-station-Garni-2055

https://github.com/SP3WRO/BresserWeatherSensorBasicAPRS

Wyjaśnione niejasności,

pobrana czysta instalacja Arduino i przetestowane wszystko po kolei.

BresserWeatherSensorReceiver by Matthias Prinke ...
<matthias-bs@web.de>

0.39.1 installed

Bresser 5-in-1/6-in-1/7-in-1 868 MHz Weather Sensor Radio Receiver for Arduino based on CC1101, SX1276/RFM95W, SX1262 or LR1121. Bresser 5-in-1/6-in-1/7-in-1 868 MHz Weather Sensor Radio Receiver...
More info

0.39.1 REMOVE

LIBRARY MANAGER

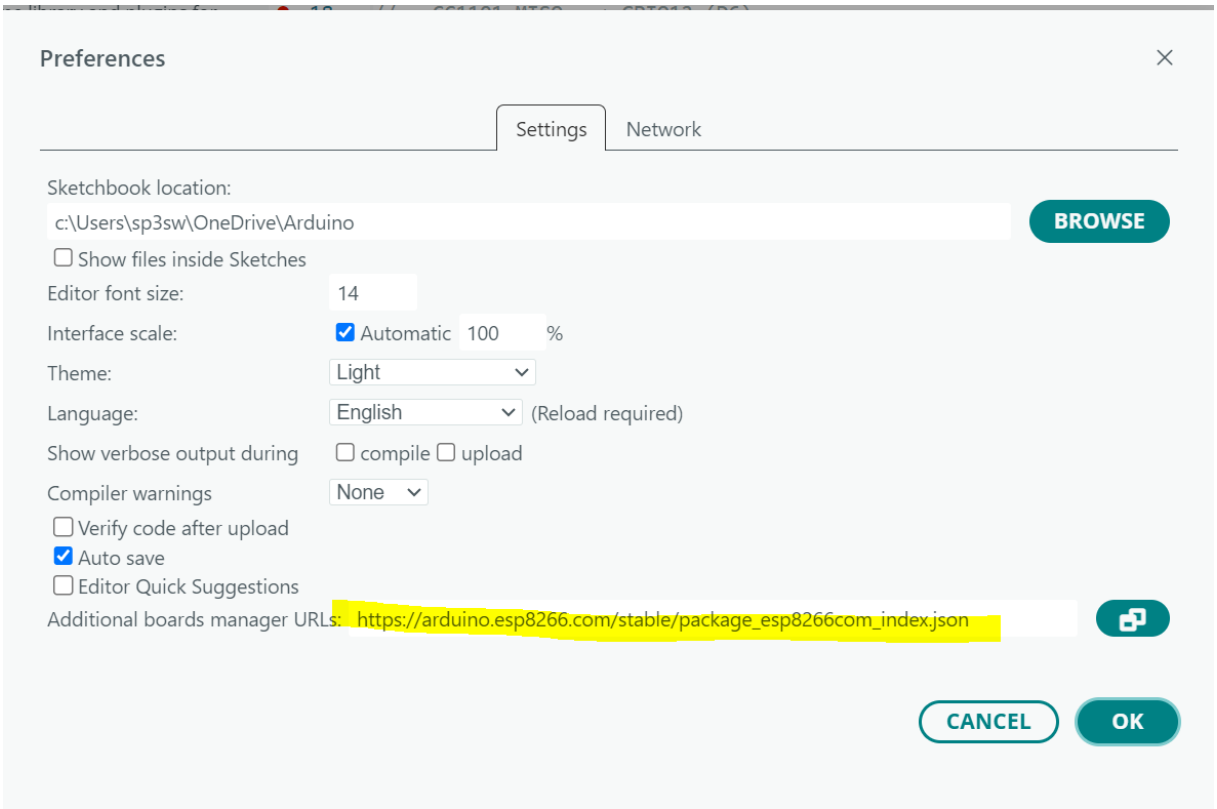
adafruit bme280

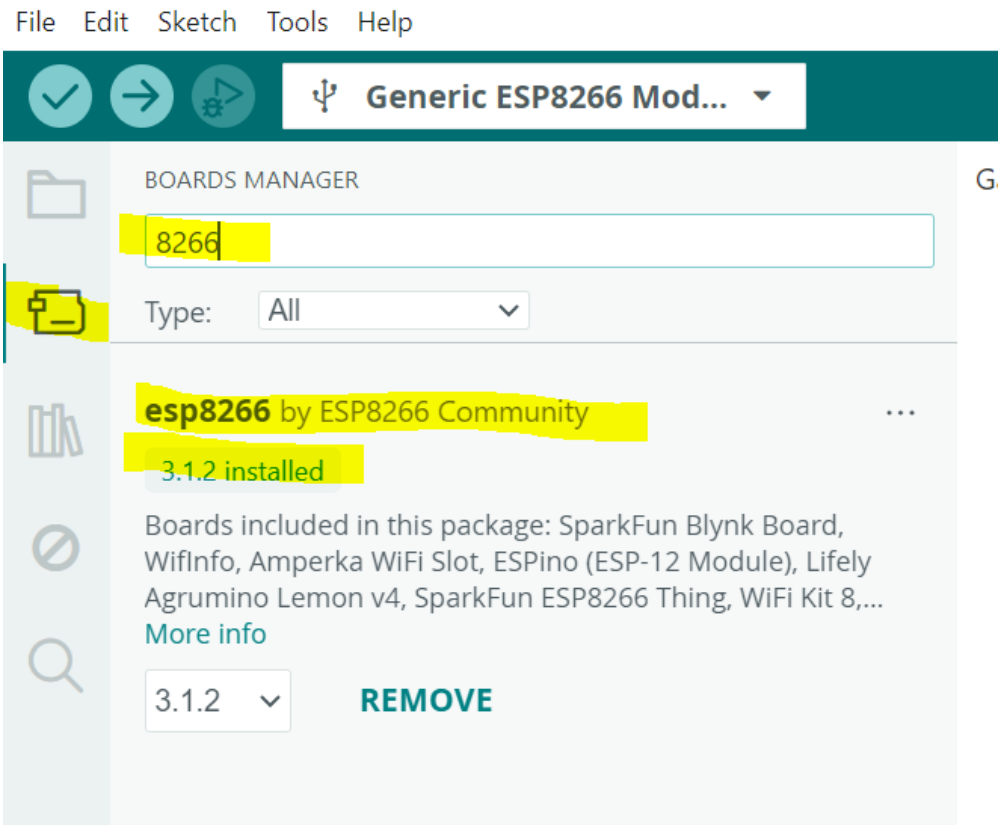
Type: All
Topic: All

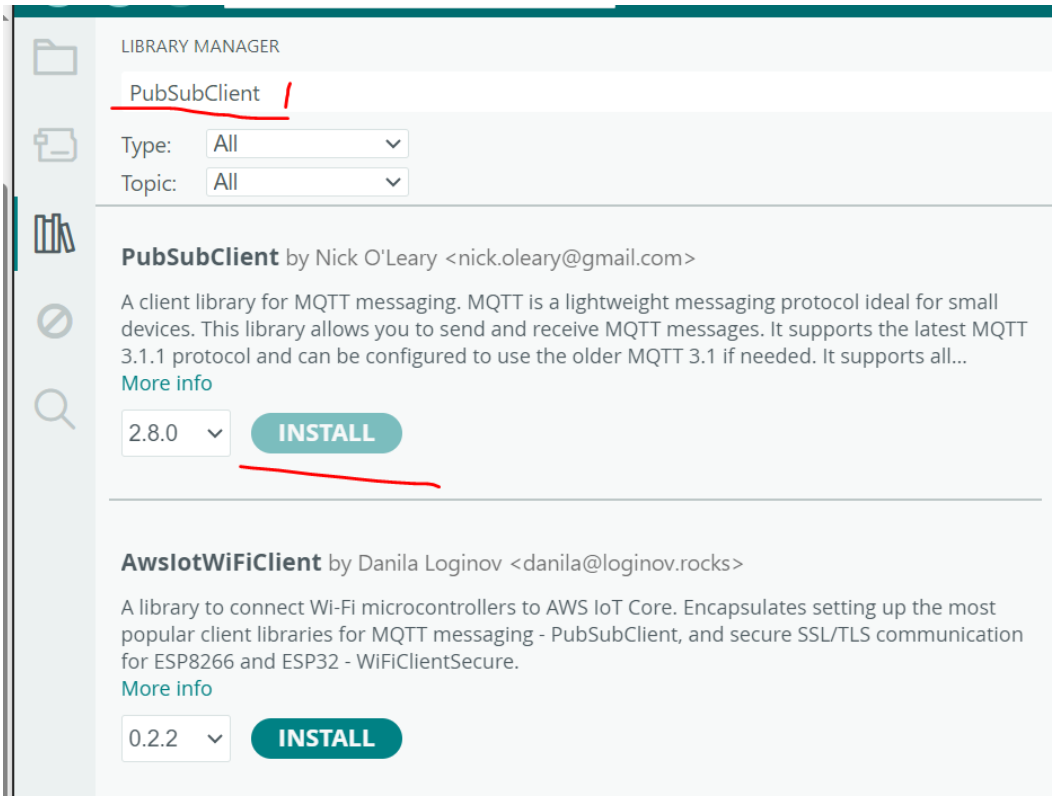
Adafruit BME280 Library by Adafruit

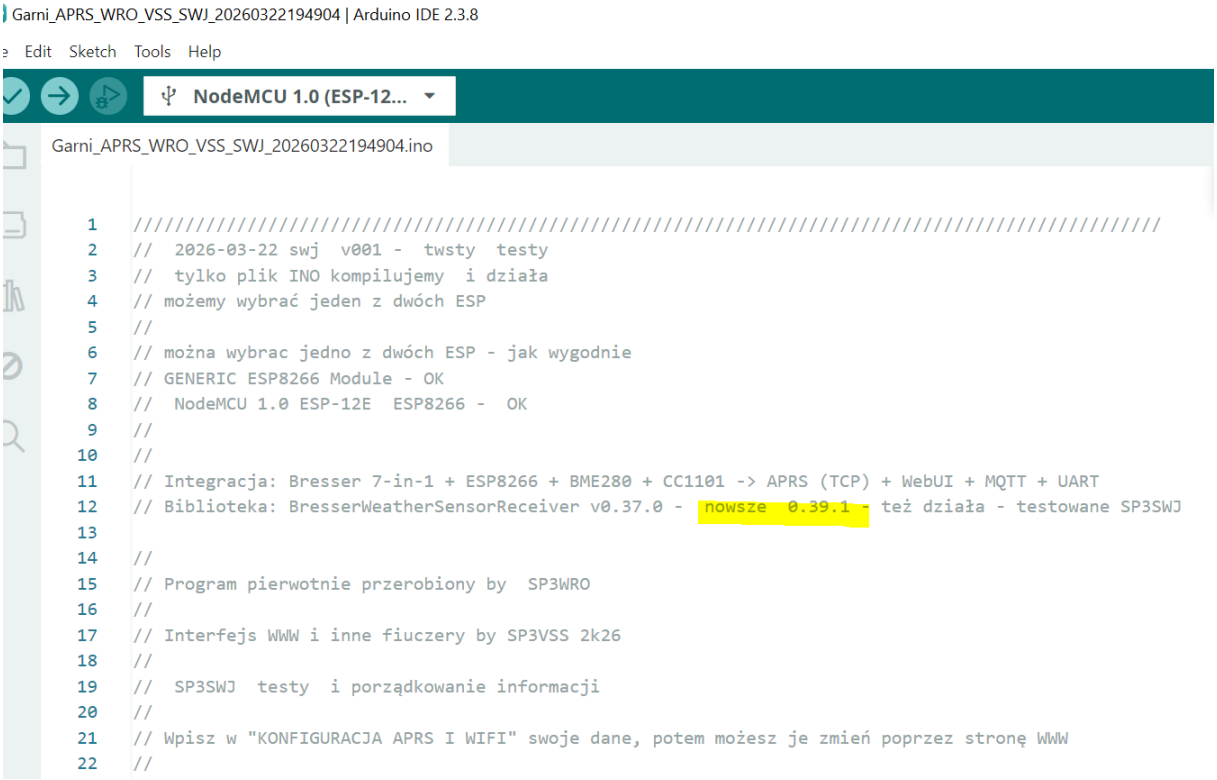
Arduino library for BME280 sensors. Arduino library for BME280 humidity and pressure sensors.
More info

2.3.0 INSTALL


Preferences
Settings
Network
Sketchbook location:
c:\Users\sp3sw\OneDrive\Arduino
BROWSE
Show files inside Sketches
Editor font size:
14
Interface scale:
Automatic
100
%
Theme:
Light
Language:
English
(Reload required)
Show verbose output during
compile
upload
Compiler warnings
None
Verify code after upload
Auto save
Editor Quick Suggestions
Additional boards manager URLs:
https://arduino.esp8266.com/stable/package_esp8266com_index.json
CANCEL
OK


File
Edit
Sketch
Tools
Help
Generic ESP8266 Mod...
BOARDS MANAGER
8266
Type:
All
esp8266 by ESP8266 Community
3.1.2 installed
Boards included in this package: SparkFun Blynk Board, WifInfo, Amperka WiFi Slot, ESPino (ESP-12 Module), Lifely Agrumino Lemon v4, SparkFun ESP8266 Thing, WiFi Kit 8,...
More info
3.1.2
REMOVE


LIBRARY MANAGER
PubSubClient
Type: All
Topic: All
PubSubClient by Nick O'Leary <nick.oleary@gmail.com>
A client library for MQTT messaging. MQTT is a lightweight messaging protocol ideal for small devices. This library allows you to send and receive MQTT messages. It supports the latest MQTT 3.1.1 protocol and can be configured to use the older MQTT 3.1 if needed. It supports all...
More info
2.8.0
INSTALL
AwsIotWiFiClient by Danila Loginov <danila@loginov.rocks>
A library to connect Wi-Fi microcontrollers to AWS IoT Core. Encapsulates setting up the most popular client libraries for MQTT messaging - PubSubClient, and secure SSL/TLS communication for ESP8266 and ESP32 - WiFiClientSecure.
More info
0.2.2
INSTALL


Garni_APRS_WRO_VSS_SWJ_20260322194904 | Arduino IDE 2.3.8
Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
Garni_APRS_WRO_VSS_SWJ_20260322194904.ino
//
// 2026-03-22 swj v001 - twsty testy
// tylko plik INO kompilujemy i działa
// możemy wybrać jeden z dwóch ESP
//
// można wybrac jedno z dwóch ESP - jak wygodnie
// GENERIC ESP8266 Module - OK
// NodeMCU 1.0 ESP-12E ESP8266 - OK
//
//
// Integracja: Bresser 7-in-1 + ESP8266 + BME280 + CC1101 -> APRS (TCP) + WebUI + MQTT + UART
// Biblioteka: BresserWeatherSensorReceiver v0.37.0 - nowsze 0.39.1 - też działa - testowane SP3SWJ
//
// Program pierwotnie przerobiony by SP3WRO
//
// Interfejs WWW i inne fiuczery by SP3VSS 2k26
//
// SP3SWJ testy i porządkowanie informacji
//
// Wpisz w "KONFIGURACJA APRS I WIFI" swoje dane, potem możesz je zmień poprzez stronę WWW
//

Output

```
esptool.py v3.0
Serial port COM11
Connecting....
Chip is ESP8266EX
Features: WiFi
Crystal is 26MHz
MAC: ec:fa:bc:c6:3e:ee
Uploading stub...
Running stub...
Stub running...
Configuring flash size...
Auto-detected Flash size: 4MB
Flash params set to 0x0340
Compressed 397520 bytes to 282591...
Writing at 0x00000000... (5 %)
Writing at 0x00004000... (11 %)
Writing at 0x00008000... (16 %)
Writing at 0x0000c000... (22 %)
Writing at 0x00010000... (27 %)
Writing at 0x00014000... (33 %)
Writing at 0x00018000... (38 %)
Writing at 0x0001c000... (44 %)
Writing at 0x00020000... (50 %)
Writing at 0x00024000... (55 %)
Writing at 0x00028000... (61 %)
Writing at 0x0002c000... (66 %)
Writing at 0x00030000... (72 %)
Writing at 0x00034000... (77 %)
Writing at 0x00038000... (83 %)
Writing at 0x0003c000... (88 %)
Writing at 0x00040000... (94 %)
Writing at 0x00044000... (100 %)
Wrote 397520 bytes (282591 compressed) at 0x00000000 in 25.0 seconds (effective 127.4 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
```

Uploading...

CANCEL

Ln 26, Col 33 Generic ESP8266 Module on COM11 2

Serial Monitor ×

Message (Enter to send message to 'Generic ESP8266 Module' on 'COM11') | New Line | 115200 baud

```
[UART] WX,17:11,22-03-2026,24.1,32,208.0,0.0,0.0,1.2,0.0,0.2,-66,1,24.7,29.4,1008.8,1018.7
[RADIO] OK | RSSI: -69.50
rl□l□r□$□□n□l□b|□□□□□□r□b□b□□nn□lnn□bb□p□$b□lrlp□n□□□l□□bn□n□□□b□□nn'l□□l`□□□□nnl`□□□nr□□□nbl□`□p□n□□r□□□b□□bn□n□b□□n

=== START RodosWX_2 ===
[CFG] UTC_OFFSET = 1
[BME] OK 0x76
[CC1101] OK
[WiFi] Laczenie........... OK
[WiFi] IP: 192.168.1.55
[NTP] Synchronizacja. OK
[NTP] Czas UTC: 1774195895
[HTTP] OK
=== SETUP OK ===

[RADIO] OK | RSSI: -63.50
[UART] WX,17:11,22-03-2026,24.1,32,208.0,0.0,0.0,1.2,0.0,0.2,-63,1,24.8,29.6,1008.9,1018.8
[UART] WX,17:11,22-03-2026,24.1,32,208.0,0.0,0.0,1.2,0.0,0.2,-63,1,24.8,29.5,1008.9,1018.8
[RADIO] OK | RSSI: -64.00
[UART] WX,17:11,22-03-2026,24.1,33,208.0,0.0,0.0,1.2,0.0,0.2,-64,1,24.8,29.4,1008.8,1018.7
[UART] WX,17:11,22-03-2026,24.1,33,208.0,0.0,0.0,1.2,0.0,0.2,-64,1,24.8,29.3,1008.8,1018.7
[UART] WX,17:12,22-03-2026,24.1,33,208.0,0.0,0.0,1.2,0.0,0.2,-64,1,24.8,29.4,1008.8,1018.7
[RADIO] OK | RSSI: -64.00
```

Ln 513, Col 72 Generic ESP8266 Module on COM11

Stacja Pogodowa APRS - SP3SWJ-13

Zobacz stację na APRS.fi

Czas lokalny:	17:00	UTC+1
Data:	22-03-2026	

Dane z Radia (Bresser 7-in-1)

Temperatura:	24.1 °C
Wilgotność:	31 %
Kierunek wiatru:	128 °
Prędkość wiatru:	0.0 m/s
Porywy wiatru:	0.0 m/s
Opady (suma):	1.2 mm
Indeks UV:	0.0
Światło:	0.1 klx
Siła sygnału CC1101:	-75 dBm
Bateria stacji:	OK

Dane z BME280 (GPIO0/GPIO3)

Temperatura:	24.5 °C
Wilgotność:	28.3 %
Ciśnienie absolutne:	1008.8 hPa
Ciśnienie QNH (MSL):	1018.7 hPa

Ostatnia aktualizacja: 11 s temu

UART TX (GPIO2/D4, 9600, co 5s)

```
WX,17:00,22-03-2026,24.1,31,128.0,0.0,0.0,1.2,0.0,0.1,-75,1,24.5,28.3,1008.8,101
8.7
```

Konfiguracja Stacji

Nazwa WiFi (SSID):
Orange_Swiatlowod_9FF0B

Hasło WiFi:
••••••••••••

Znak wywoławczy APRS:
SP3SWJ-13

Passcode APRS:
14580

Szerokość geograficzna:
52,943400

Długość geograficzna:
16,123400

Wysokość stacji n.p.m. [m]:
85,0

Interwał raportów APRS (min):
15

Strefa czasowa (UART)

Przesunięcie względem UTC (godziny, -12 do +14):
UTC+1

⚠ Czas APRS zawsze wysyłany jako UTC (standard APRS).
Strefa czasowa dotyczy tylko ramek UART i wyświetlania na stronie.

MQTT 3.1.1

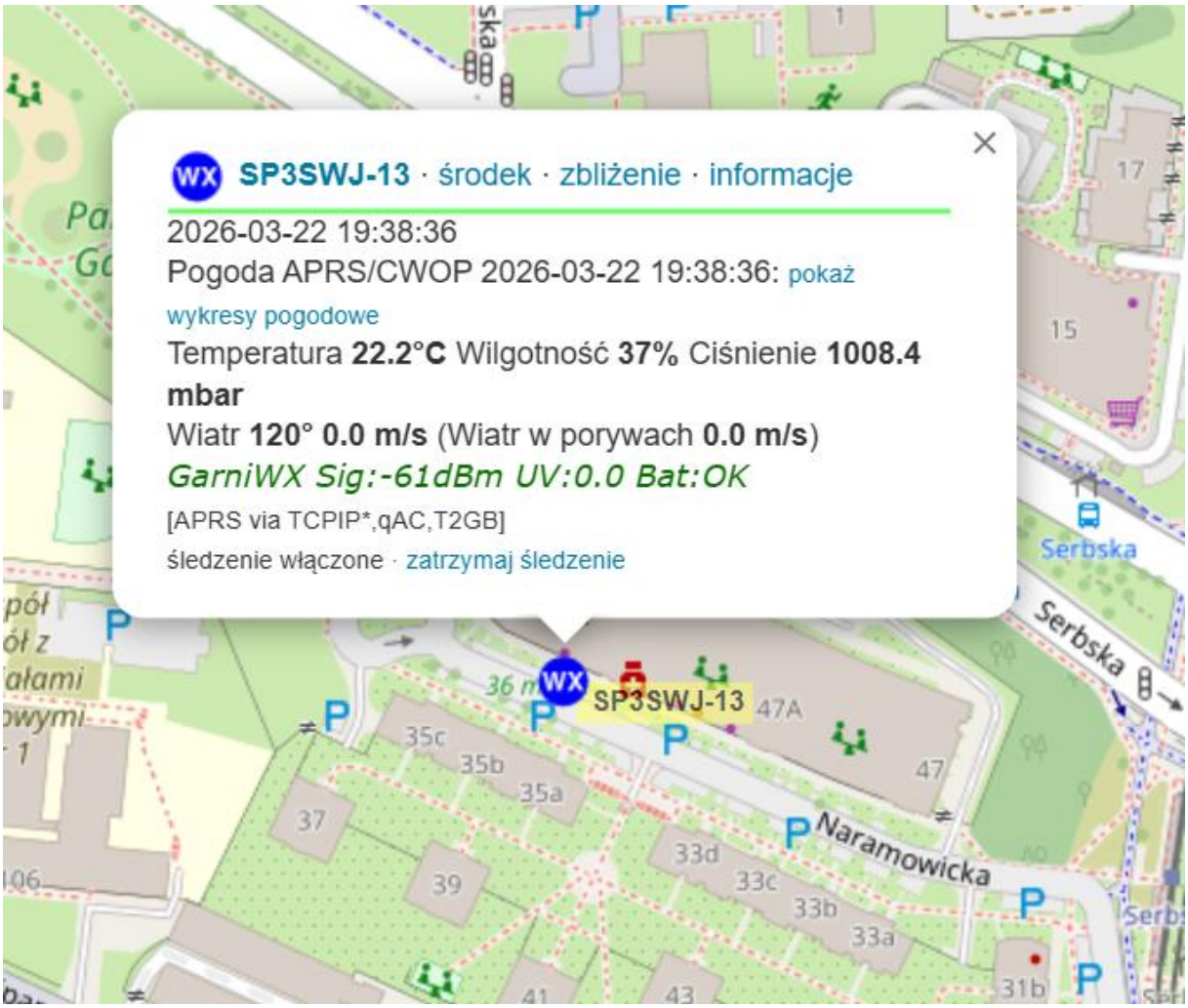
MQTT włączony:
Nie

MQTT Host:
broker.hivemq.com

MQTT Port:
1883

MQTT User:

MQTT Hasło:


SP3SWJ-13 · środek · zbliżenie · informacje
2026-03-22 19:38:36
Pogoda APRS/CWOP 2026-03-22 19:38:36: pokaż wykresy pogodowe
Temperatura 22.2°C Wilgotność 37% Ciśnienie 1008.4 mbar
Wiatr 120° 0.0 m/s (Wiatr w porywach 0.0 m/s)
GarniWX Sig:-61dBm UV:0.0 Bat:OK
[APRS via TCPIP*,qAC,T2GB]
śledzenie włączone · zatrzymaj śledzenie
SP3SWJ-13
Serbska
Naramowicka
47A
47
35c
35b
35a
37
39
33d
33c
33b
33a
31b
17
15

//end